KnowledgeGraphEval 2026: The First Shared Task on Arabic Knowledge Graphs

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https://sina.birzeit.edu/KnowledgeGraphEval

Abstract

We present KnowledgeGraphEval 2026, the first Arabic shared task dedicated to knowledge graph (KG) construction from text. The task focuses on extracting named entities and semantic relations from Arabic text to build structured, machine-readable knowledge representations. It includes two subtasks: (*i*) **AdaptNER**, which evaluates cross-domain named entity recognition (NER), and (*ii*) **Relation Extraction (RE)**, which predicts semantic relations between predefined entity pairs. For AdaptNER, we release the MSA subset of *Konooz*, an Arabic NER benchmark annotated with 21 entity types across 10 domains. For RE, we release *Wojood*^{Relations}, an Arabic RE dataset annotated with 40 relation types. At test time, 14 teams participated, of which 13 submitted system description papers; and one RE paper was subsequently rejected. The best systems achieve micro-F1 scores of 75.96% and 94.06% on AdaptNER and RE, respectively. Results show that cross-domain NER remains challenging, while RE systems continue to face difficulties with rare and semantically confusable relations. We release the datasets, baselines, and evaluation tools for reproducible research in Arabic information extraction and KG construction.

1 Introduction

Knowledge graphs (KGs) have become a fundamental component of modern AI and NLP systems. A KG represents knowledge as entities and the semantic relations between them in a structured, machine-readable form. This representation supports a wide range of downstream applications, including question answering (Shaheen and Ezzeldin, 2014), semantic search (Guha et al., 2003), information extraction and retrieval (Jiang et al., 2016), word sense disambiguation (Jarrar et al., 2023b; Al-Hajj and Jarrar, 2021), language understanding (Khalilia et al., 2024), interoperability (Jarrar et al., 2011), and cybersecurity (Tikhomirov et al., 2020). More recently, KGs have become integral to Retrieval-Augmented Generation (RAG) systems for Large Language Models (LLMs), where structured knowledge grounds and constrains model outputs (Zhu et al., 2025).

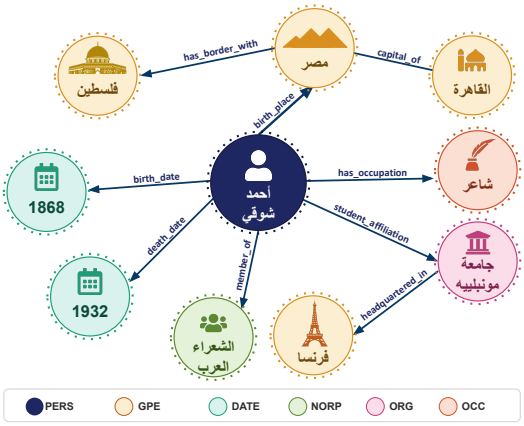


Figure 1: An example of a Knowledge Graph illustrating extracted entities and their relations.

Constructing KGs from text requires identifying both the entities and the semantic relations that connect them. This process primarily involves two tasks: (*i*) *Named Entity Recognition* (NER), which identifies entity mentions and assigns them predefined types such as `PERS`, `ORG`, and `LOC`; and (*ii*) *Relation Extraction* (RE), which identifies the semantic relations between these entities. For example, in Figure 1, الشاعر is recognized as `OCC` and أحمد شوقي as `PERS`, with `has_occupation` identifying the relation between them.

Both tasks present substantial challenges for Arabic. For NER, existing systems and datasets are predominantly focused on narrow domains, particularly news. Their performance can degrade under shifts in domains, dialects, and writing styles due to changes in entity distributions and vocabulary (Hamad et al., 2025). Arabic RE is similarly under-resourced, with limited availability of large-

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scale datasets covering diverse and fine-grained relation types. This resource gap hinders the development of robust Arabic KG construction systems and leaves end-to-end extraction from diverse, multi-domain text comparatively underexplored.

To address this gap, we introduce **KnowledgeGraphEval** 2026, the first ArabicNLP shared task dedicated to knowledge graph construction. The shared task focuses on extracting entities and semantic relations from Arabic text to build structured knowledge representations. It comprises two subtasks: (*i*) **Cross-Domain Named Entity Recognition (AdaptNER)**, which evaluates domain-adaptive NER, and (*ii*) **Relation Extraction (RE)**, which identifies semantic relations between predefined entity pairs. For AdaptNER, we release ***Konooz***, a new Arabic NER benchmark covering 21 entity types across 10 domains (Hamad et al., 2025). For RE, we release ***Wojood***^{Relations}, an Arabic dataset annotated with 40 relation types (Aljabari et al., 2025b).

Teams were invited to explore recent modeling strategies, including pretrained language models, large language models, parameter-efficient fine-tuning, retrieval-augmented methods, and ensemble-based approaches. A total of 44 unique teams registered for the shared task. Of these, eight teams submitted models for evaluation on AdaptNER, while nine teams submitted models for evaluation on RE.

2 Related Work

2.1 Domain Adaptation in NER

Most Arabic NER corpora are in-domain, focusing primarily on MSA, including *ANERCorp* (Benajiba et al., 2007), *OntoNotes* 5.0 (Weischedel et al., 2013), and *Wojood* (Jarrar et al., 2022), along with its extensions *Wojood*$_{Fine}$ (Liqreina et al., 2023; Jarrar et al., 2023a) and *Wojood*^{Gaza} (Jarrar et al., 2024). As a result, out-of-domain benchmarks for Arabic NER remain limited, hindering systematic evaluation of cross-domain generalization. *Konooz* addresses this gap by providing a benchmark spanning domains and dialects distinct from the *Wojood* training distribution (Hamad et al., 2025).

Domain adaptation for NER has been extensively investigated with pretrained language models, including approaches based on self-training, knowledge distillation, adversarial training, and adaptive pre-training (Zhou et al., 2023; Ge et al., 2023; Han and Eisenstein, 2019). For Arabic, however, existing studies have primarily examined MSA-to-dialect adaptation and have considered relatively small entity-type inventories Khalifa et al. (2021); Mekki et al. (2022). Thus, the combination of MSA domain shift and broader entity-type coverage remains underexplored, which we address by introducing AdaptNER based on *Konooz*.

2.2 Relation Extraction

Arabic relation extraction remains substantially under-resourced, with existing datasets offering limited coverage in terms of domains, relation types, and annotation scale (Doddington et al., 2004; Seganti et al., 2021; Huguet Cabot et al., 2023). This scarcity has constrained the systematic development and evaluation of Arabic RE systems. Consequently, prior work has largely relied on cross-lingual transfer (Lan et al., 2020) or task-specific approaches, including evidence-augmented methods (Rakan Al Mraikhat et al., 2024), and joint neural extraction frameworks (El Khbir et al., 2022; Osman et al., 2023). Differences in datasets, relation inventories, and evaluation settings further limit direct comparison across studies (Aljabari et al., 2024).

Wojood^{Relations} (Aljabari et al., 2025b) addresses this resource gap by providing a large-scale, manually annotated Arabic RE corpus with fine-grained relation types and formally defined ontology constraints. Building on *Wojood*^{Relations}, we establish the RE subtask to provide a unified evaluation setting and enable systematic comparison of different approaches on a common benchmark.

2.3 Shared Tasks in Knowledge Graphs

Several previous shared tasks have supported progress in KG construction and enrichment. The Open Knowledge Extraction (OKE) challenges evaluated entity recognition, linking, typing, and knowledge-base enrichment from text (Nuzzolese et al., 2015, 2016). SemTab focuses on semantic annotation and linking of tabular data to KGs (Jiménez-Ruiz et al., 2020), while the OAEI KG track evaluates instance- and schema-level alignment between KGs (Hertling and Paulheim, 2020). WikiKG90M, introduced within the OGB Large-Scale Challenge, benchmarks large-scale KG completion through link prediction (Hu et al., 2021).

For Arabic, previous shared tasks have mainly focused on individual extraction components. WojoodNER 2023 and 2024 introduced benchmarks for flat, nested, and fine-grained Arabic NER (Jar-

rar et al., 2023a, 2024). KnowledgeGraphEval 2026 extends this direction toward KG construction by combining cross-domain NER with fine-grained semantic relation extraction.

3 Shared Task Overview

3.1 Subtask 1: AdaptNER Cross-Domain Named Entity Recognition

NER is a fundamental NLP task that identifies and classifies named entities in text into predefined categories (see Figure 2). This subtask targets cross-domain adaptation for Arabic NER. Participants develop NER systems trained on *Wojood* and evaluated on the MSA subset of the existing *Konooz* benchmark, which spans ten unseen target domains, to measure how well NER models generalize across Arabic domains. Systems must be trained using only the training and development splits of the *Wojood* NER corpus as the source domain.

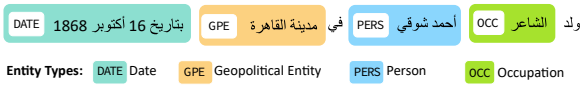


Figure 2: Example of NER annotation.

Participants are encouraged to go beyond standard supervised fine-tuning and apply domain adaptation methods such as self-training, adversarial training, or feature alignment to improve robustness and reduce the performance gap between source and target domains. Submissions will be ranked by their aggregate NER performance (Micro-F1) across the ten target domains.

3.2 Subtask 2: Relation Extraction

Relation extraction is central to knowledge graph construction, as it identifies the semantic connections that form edges between entities (see Figure 3). Building on this role, Subtask 2 focuses on extracting such relations from Arabic text. Each input sentence is provided with a single subject–object entity pair, specified by their entity spans. Participants must predict the semantic relation between the pair from a predefined set of relation types, or assign *no_relation* when no predefined relation holds between the entities.



Figure 3: Example of sentence-level relation extraction annotation.

3.3 Restrictions

This section outlines the guidelines for participating in KnowledgeGraphEval 2026. These rules have been put in place to ensure fairness and transparency for all participants.

Modeling. For Subtask 1 (AdaptNER), participants are only allowed to use encoder-based models fine-tuned on *Wojood*, using *Konooz* solely for evaluation. Training on *Konooz* is not permitted. For Subtask 2 (RE), participants may also use LLMs for fine-tuning or in-context learning, in addition to encoder-based models fine-tuned on $Wojood^{Relations}$.

External data. In both subtasks, participants may use external data or resources for additional training or as information sources during inference. Any use of external data must be explicitly reported in the system description paper, including how they were used.

4 Subtask 1: Cross-Domain Named Entity Recognition (AdaptNER)

4.1 Dataset

For Subtask 1, we use *Konooz* (Hamad et al., 2025), the first Arabic NER benchmark corpus spanning ten domains and 16 dialects, including Modern Standard Arabic (MSA). The MSA portion contains approximately 8K tokens per domain. For the shared task, we released an MSA subset referred to as " *Konooz*-MSA ", covering all ten domains, with 5K tokens per domain, comprising 50K tokens in total. The ten MSA domains are Politics, Economics, Finance, History, Law, Science, Health, Agriculture, Art, and Sport. All domains were held out exclusively for evaluation. Table 1 presents the number of sentences available in each domain for participant evaluation.

Participants were also provided with the training and development splits of *Wojood* as labeled source-domain data, comprising approximately 448K tokens and 72K entity mentions across 21 entity types. These splits were the only labeled data permitted for training the models.

We selected a subset of the *Konooz*-MSA test set as a development set -*Konooz*-dev- to support participants in evaluating their models during target-domain adaptation and model tuning, while keeping its gold annotations hidden. For the final evaluation, participants' systems were evaluated on the full *Konooz*-MSA test set, which included the "*Konooz*-dev" subset.

Domain	Sentences	Domain	Sentences
Agriculture	176	History	130
Art	130	Law	129
Economics	132	Politics	128
Finance	237	Science	149
Health	281	Sport	153
Total		**1,645**	

Table 1: Number of sentences per domain in the *Konooz*-MSA test set.

Format The released data follows a column-based BIO format: each line contains a token followed by 21 BIO-tag columns, one per entity type (`CARDINAL`, `CURR`, `DATE`, `EVENT`, `FAC`, `GPE`, `LANGUAGE`, `LAW`, `LOC`, `MONEY`, `NORP`, `OCC`, `ORDINAL`, `ORG`, `PERCENT`, `PERS`, `PRODUCT`, `QUANTITY`, `TIME`, `UNIT`, `WEBSITE`). Sentences are separated by a blank line, and the ten domains are concatenated into a single file in alphabetical order (Table 1).

4.2 Baselines

As a baseline for this subtask, we provide a single no-adaptation baseline, in which an AraBERT (Antoun et al., 2020) model is trained exclusively on *Wojood* and evaluated directly on *Konooz*-MSA without applying any domain adaptation technique. This baseline reflects the performance of a NER model when deployed on unseen target domains without adaptation. The implementation is publicly available on GitHub[1].

4.3 System Descriptions

Table 2 summarizes the techniques employed by the participating teams in Subtask 1. According to the shared task rules, participants were not allowed to use the unlabeled *Konooz*-MSA data for training. However, some participants leveraged the unlabeled data for pre-training. Therefore, we divided the subtask into two tracks. In Track 1, participants followed the shared task rules and did not use *Konooz*-MSA data during training. In Track 2, participants used the unlabeled *Konooz*-MSA data for pre-training. In both tracks, final evaluation is conducted on unlabeled *Konooz*-MSA test data.

NAMAA(Fathallah et al., 2026) NAMAA combines three complementary models for nested NER: two BIO taggers, using ARBERTv2 (Abdul-Mageed et al., 2021) and XLM-R-large (Conneau et al., 2020) encoders with learned mixtures of their top four hidden layers and 21 per-type constrained CRF decoders, plus a third model pairing ARBERTv2 with GlobalPointer to predict typed spans directly for explicit boundary modeling. An entity is retained only when at least two models agree exactly on span and type, yielding a conservative, high-precision ensemble.

Wasl (Abdo and Adam, 2026) Wasl fine-tunes AraBERTv02 with 21 independent per-type BIO heads over a shared encoder, matching the 21-column format directly. Each head trains with a compound loss (cross-entropy, Dice, Tversky, Focal) plus a variance penalty to keep frequent types from dominating rare ones. To handle domain-shift variance, Wasl ensembles three seeds via span-level majority vote — chosen over mean-probability or union combination, since seed disagreement on the blind set proved systematic rather than random, favoring a precision-preserving vote.

PTUK-NLP (Duridi et al., 2026) PTUK-NLP follows strict zero-shot transfer, pairing ArBERT with a nested tagger of 21 independent BIO heads. Label imbalance is addressed by oversampling sentences with rare entity types and replacing eligible non-nested mentions with same-type mentions from a Wojood-derived gazetteer, preserving token alignment and annotations. At inference, lightweight rules extend fragmented website spans and recognize unseen Latin-script currency codes. *Konooz*-MSA text is never used—for pretraining, fine-tuning, model selection, or hyperparameter optimization—making this a simple, data-centric alternative to target-aware adaptation.

McMasterNLP (Serour et al., 2026) McMasterNLP builds a heterogeneous mixture of token- and span-level experts: token classifiers on AraBERTv2, CAMeLBERT-Mix, and MARBERTv2, plus an AraBERTv2 span classifier, trained on data augmented by replacing entity mentions with same-type mentions from other sentences. A fifth AraBERTv2 expert, adapted via masked-language-model pretraining on unlabeled

[1] https://github.com/SinaLab/ArabicNER

Team	Backbone						Classification			Domain Adaptation						Method Summary
	AraBERTv2	ARBERTv2	MARBERT	XLM-R	CAMeLBERT	LLM	Multi-head BIO	CRF	GlobalPointer	Ensemble	DAPT/TAPT	Self-training	DANN/Advers.	SWA	Data Aug.	
NAMAA	✓	✓		✓				✓	✓	✓						Two BIO-CRF taggers (ARBERTv2, XLM-R) plus an ARBERTv2 GP classifier, combined via ≥2/3 exact-span voting for a high-precision.
Wasl	✓						✓			✓						AraBERTv02 with 21 independent per-type BIO heads, trained with a compound loss (CE + Dice + Tversky + Focal plus a variance penalty). Three random seeds are then combined by span-level majority vote
McMasterNLP	✓		✓		✓	✓	✓			✓	✓				✓	Token- and span-level ensemble (AraBERTv2, CAMeLBERT, MARBERTv2) with a Konooz-adapted tie-breaker; Qwen3-14B adjudicates unresolved spans.
PTUK-NLP	✓	✓					✓								✓	Strict zero-shot ArBERT nested tagger (21 heads); oversampling and gazetteer-based mention replacement address label imbalance, with inference-time rules for WEBSITE spans and currency codes.
ArmyForNER		✓					✓			✓	✓	✓				ARBERTv2, adapted via MLM pretraining on *Konooz*; a multi-seed ensemble generates self-training pseudo-labels, combined via soft voting.
AdaptLinker	✓						✓					✓	✓	✓		AraBERT is adapted via GRL domain classification and confidence-gated pseudo-labels, with final weights from SWA over the last four checkpoints.
YafaNER	✓						✓			✓	✓	✓			✓	AraBERTv2 with TAPT on *Konooz*; three teachers generate filtered soft labels, distilled into students combined via entity-specific F1 weights.
ANLP_UniSo	✓		✓						✓	✓	✓	✓				MARBERT + GP, adapted via DAPT and self-training on Konooz, then stacked with AraBERT using entity-specific decoding thresholds.

Table 2: Overview of Subtask 1 (AdaptNER) systems: backbone family, classification, and domain adaptation.

Konooz-MSA text, breaks ties during voting. Spans supported by multiple encoder families are retained; unresolved candidates go to Qwen3-14B for constrained multiple-choice adjudication, which cannot alter already accepted predictions.

ArmyForNER (Syulistyo et al., 2026) A multi-stage ARBERTv2 pipeline: the encoder is first adapted to the target domains via masked-language-model pretraining on unlabeled *Konooz*-MSA sentences, then fine-tuned on Wojood with 21 parallel BIO heads, allowing overlapping entity types to be predicted independently. An ensemble trained with varied adaptation settings and seeds generates pseudo-labels for *Konooz*-MSA, which are combined with Wojood for self-training; three resulting models are then combined via token-level soft voting.

AdaptLinker (Ho and Thin, 2026) AdaptLinker uses a shared AraBERT encoder with 21 independent BIO heads for nested NER. Training begins with supervised fine-tuning on Wojood, then continues with joint source-target adaptation on labeled Wojood and unlabeled *Konooz*-MSA sentences: a gradient-reversal domain classifier encourages domain-invariant representations, while a frozen teacher generates confidence-thresholded target-domain pseudo-labels. Final parameters come from stochastic weight averaging over the last four adaptation checkpoints.

YafaNER (Hamed et al., 2026) YafaNER builds an AraBERTv2 multi-head BIO system combining several adaptation techniques: task-adaptive pretraining on unlabeled *Konooz*-MSA text, then three R-Drop-regularized teacher models fine-tuned on Wojood (one with strong augmentation for diversity). Teachers generate soft labels for *Konooz*, filtered by confidence and entropy thresholds; three student models then train jointly on gold Wojood annotations and these filtered soft labels, with predictions combined via entity-specific weights from per-type validation F1 scores.

ANLP_UniSo (Bergaoui et al., 2026) Pairs MARBERT with a GlobalPointer decoder (Rotary Position Embeddings) to predict spans directly, capturing overlapping and nested entities without flattening. MARBERT undergoes continued MLM pretraining on unlabeled *Konooz*, then fine-tunes on Wojood; a pseudo-labeling stage then uses this model to label *Konooz*, merged with Wojood for self-training. A second AraBERT-based model is combined with MARBERT's via stacking, and entity-specific decoding thresholds are tuned to balance per-category precision and recall.

4.4 Evaluation

For Subtask 1, participants submit their predictions through the official Codabench competition page.[2] The submission consists of one `.txt` file containing predictions for all ten domains. The file follows the CoNLL format: each line contains a token followed by its IOB2 tag (Sang and Veenstra, 1999), separated by a space, with sentences separated by

[2] https://www.codabench.org/competitions/17349/

blank lines. Predictions for the ten domains must be concatenated in the domain order specified on Codabench. Evaluation uses strict entity-level matching: a predicted entity is correct only if both its span boundaries and entity type match the gold annotation. The primary metric is micro-averaged F1 across all 21 entity types and ten domains.

4.5 Results and Discussion

Table 3 presents the official results of AdaptNER. In track 1, NAMAA achieved the highest F1 of 75.96, followed by Wasl and PTUK-NLP. All described systems exceeded the unadapted AraBERTv02 baseline, with absolute improvements ranging from 6.0 to 12.0 points. In track 2, some participants utilized the provided unlabeled "*Konooz*-dev" for target-domain adaptation using adversarial training, while others used the blind *Konooz*-MSAtest data during training.

Rank	Team	Precision	Recall	Micro-F1	Macro-F1
Track 1					
1	NAMAA	**83.83**	69.44	**75.96**	**66.00**
2	Wasl	80.60	69.82	74.83	63.16
3	McMasterNLP	75.00	**74.00**	74.38	65.00
4	PTUK-NLP	71.84	66.59	69.11	60.70
Track 2					
1	ArmyForNER	82.11	68.77	74.85	65.20
2	AdaptLinker	75.48	71.21	73.29	63.37
3	YafaNER	73.25	69.52	71.33	60.91
4	ANLP-UniSo	81.43	63.04	71.06	61.01
–	Baseline	63.87	62.30	63.00	44.00

Table 3: Official results for Subtask 1 (AdaptNER) (%).

A common theme across the participating systems is ensembling pretrained language models, either by combining different Arabic or multilingual encoders, including AraBERT, ARBERT, MARBERT, CAMeLBERT, and XLM-R, or by combining multiple instances of the same model trained with different random seeds. NAMAA (75.96) ensembles three pretrained Arabic language models, while Wasl (74.83) ensembles a single pretrained model across multiple seeds. Their results indicate that model diversity, either across architectures or random initializations, can provide strong performance for cross-domain adaptation without incorporating *Konooz*-MSAduring training.

Other systems employ continued pretraining through Domain-Adaptive Pretraining (DAPT) or Task-Adaptive Pretraining (TAPT). McMasterNLP (74.99) ensembles four pretrained models, including one adapted through TAPT on unlabeled *Konooz* data. ArmyForNER (74.85), YafaNER (71.33), and ANLP_UniSo (71.06) similarly employ DAPT or TAPT, combined with methods such as pseudo-labeling, knowledge distillation, or adversarial alignment. Notably, McMasterNLP, despite using *Konooz* during training through TAPT, achieves a score comparable to the aforementioned systems that did not use *Konooz* during training. This suggests that incorporating the target-domain data available to the team did not yield a clear performance advantage in this setting.

Self-training provides mixed results across the submissions. YafaNER combines adaptation with pseudo-labeling but achieves lower performance than the top-ranked systems, while AdaptLinker (73.29) combines self-training with adversarial alignment using the *Konooz*-dev development data. These results suggest that pseudo-labeling does not consistently improve cross-domain NER performance, potentially because errors in pseudo-labels can propagate during adaptation.

Data augmentation provides another approach to improving adaptation. PTUK-NLP (74.85) augments only the *Wojood* training data using two strategies — oversampling and a Wojood-derived gazetteer — to increase the number of rare entity type instances. Its performance is comparable to the other high-performing systems and substantially exceeds the baseline (+6%), suggesting that improving the representation of underrepresented entity types can benefit cross-domain NER.

Overall, the results indicate that more elaborate adaptation pipelines do not necessarily translate into higher performance. Adaptation quality appears to depend less on pipeline complexity. The strongest systems instead consistently incorporate ensemble-based strategies, either by ensembling multiple models or multiple seeds of a single model. The precision–recall patterns further suggest that these ensemble strategies favor precision, although this is accompanied by lower recall. Thus, ensemble diversity appears to be a promising component for Arabic cross-domain NER.

5 Subtask 2: Relation Extraction (RE)

5.1 Dataset

For Subtask 2, we use *Wojood*^{Relations} (Aljabari et al., 2025b), an Arabic relation extraction (RE) corpus built on the Wojood NER corpus, which contains $33K$ sentences and $550K$ tokens (Jarrar et al., 2022). *Wojood*^{Relations} contains approximately $14K$ annotated relation triples spanning 40 relation types, including organizational, geograph-

ical, and familial relations. The corpus follows a multi-relation setting in which a sentence may contain multiple entity pairs associated with distinct relations.

For the KnowledgeGraphEval 2026 shared task, we cast RE as a *relation classification* problem. Given an Arabic sentence and a target subject–object entity pair occurring in that sentence, the task is to predict the relation holding between the two entities. Each annotated relation triple in *Wojood*^{Relations} is therefore converted into an individual classification instance, while retaining the complete sentence as contextual input. Consequently, a sentence containing multiple annotated relations contributes multiple instances, each corresponding to a different target entity pair. For example, the relation between قبة الصخرة and مدينة القدس in the sentence below is represented as a single classification instance with the label `located_in`.

sentence_id: 100009219
triple_id: 8311
sentence: تقع قبة الصخرة في مدينة القدس، وقد شُيدت في عهد الخليفة عبد الملك بن مروان عام ٦٩١م.
subject: قبة الصخرة
object: مدينة القدس
relation: located_in

We further introduce negative instances to explicitly model the absence of a relation. Specifically, for entity pairs occurring in the same sentence but not associated with an annotated relation, we assign the `no_relation` label. This label is included in the training and validation data to expose systems to the distinction between related and unrelated entity pairs. The resulting classification space therefore comprises 41 labels: the 40 annotated relation types and `no_relation`. The test set, however, contains only instances corresponding to annotated relations and does not include `no_relation` instances. Accordingly, `no_relation` is excluded from the official evaluation.

The final dataset contains approximately $24K$ relation-classification instances across the training, validation, and test splits. The training set is provided with gold relation labels, while the validation and test sets are provided without gold labels and remain blind to participants. Table 4 summarizes the split statistics.

Split	Train	Validation	Test
# Instances	17,381	2,074	4,386

Table 4: *Wojood*^{Relations} split statistics for KnowledgeGraphEval 2026.

5.2 Baselines

We use the NLI-based relation extraction (NLI-RE) system introduced with the *Wojood*^{Relations} corpus as the baseline. The approach formulates relation classification as binary natural language inference: given a sentence and a predefined subject–object entity pair, each candidate relation is converted into a hypothesis using a relation-specific template, with the sentence serving as the premise. A BERT-based NLI classifier then predicts whether the candidate relation is entailed. The implementation is publicly available on GitHub.[3]

5.3 System Descriptions

Table 5 provides an overview of the approaches adopted by the participating systems. We describe these approaches below.

U4RASD (Abbas et al., 2026) U4RASD first identifies and types entity occurrences using a NeoAraBERT-based TypePredictor, then uses WojoodOntology (Aljabari et al., 2025a) domain and range constraints to determine the valid candidate relations for each entity pair. Within this pipeline, the team evaluates two approaches: encoder-based EntityPair models with typed entity markers and separate relation-existence and relation-classification heads, and QLoRA-adapted LLMs, including Yehia and Qwen3.5, formulated as either multiple-choice or sequence-classification models. The best-performing system, Yehia-RE, achieves 94.06% Micro-F1 on the official test set.

NAMAA (Fathallah et al., 2026) Three decoder-only LLMs are fine-tuned with LoRA, each receiving the Arabic sentence, ordered subject and object mentions, and the complete relation-label inventory. Rather than relying on free-form generation, inference is constrained to the 41 valid labels by scoring each candidate relation using its mean teacher-forced token log-likelihood. Predictions from the three models are combined via majority voting, with Qwen3.5-4B breaking ties when all three models disagree.

[3] https://github.com/SinaLab/ArabicNLI-RE

Team	Backbone				Ensembling				Constraints			Method Summary
	AraBERT	XLM-R	ARBERTv2	LLM	CV	Multi-model	Vote/Avg	None	Ont. (soft)	Ont. (hard)	Imb.-aware	
U4RASD				✓				✓		✓	✓	Yehia-7B fine-tuned with QLoRA as a row-specific Arabic multiple-choice task over the full vocabulary.
NAMAA				✓		✓	✓					Three LoRA-tuned causal LLMs scored by length-normalized log-likelihood over 41 labels, combined by majority vote.
ArmyForRE	✓	✓	✓			✓	✓			✓	✓	Seven encoders (focal loss) combined by soft voting; incompatible relations masked by ontology constraints.
Wasl	✓				✓		✓		✓			Typed markers on AraBERTv02 (5-fold CV, logit-averaged) over 41 relations, with a soft ontology-based type-constraint penalty at decoding.
McMasterNLP	✓				✓		✓		✓			Five-fold AraBERT ensemble for 41-way classification, reranked by a five-fold NLI model, with schema penalties.
ISNAD	✓							✓			✓	AraBERTv2 fine-tuned end-to-end with entity markers; class imbalance addressed via moderated weighted sampling.
YafaRE			✓					✓				ArBERTv2 fine-tuned as a relation classifier, then adapted via transductive self-training on pseudo-labeled test inputs.

Table 5: Overview of Subtask 2 (Relation Extraction) systems: Design choices adopted by participating systems.

ArmyForRE (Shoumi et al., 2026) ArmyForRE formulates RE as a 41-way classification task using typed entity markers and pretrained language encoders to represent subject–object pairs. Focal loss addresses class imbalance. At inference, ontology-based domain and range constraints mask incompatible relations. The final system is a seven-model soft ensemble comprising four independently seeded XLM-RoBERTa-large models and three Arabic BERT-based models, with predictions aggregated through soft voting.

Wasl (Abdo and Adam, 2026) Wasl uses a pretrained encoder with typed punctuation-based entity markers and a two-layer feed-forward classifier over the concatenated subject and object start-marker representations. The model is trained as a 41-way classifier. The final system ensembles five models trained with sentence-grouped 5-fold cross-validation, with a soft ontology-based type-constraint penalty applied during decoding.

McMasterNLP (Serour et al., 2026) McMasterNLP uses an AraBERT-based relation classifier with explicit subject and object markers, trained with sentence-grouped 5-fold cross-validation and auxiliary objectives for entity-type prediction, together with embedding-level FGM adversarial training. The classifier produces relation scores, after which candidate relations are verbalized as Arabic hypotheses and reranked using a five-fold AraBERT-based NLI ensemble. Final predictions combine the classifier and NLI probabilities with a soft entity-type schema penalty.

ISNAD (Hassaneen et al., 2026) Fine-tunes AraBERTv2 with cross-entropy loss for multi-class relation classification, using entity markers inserted around the subject and object spans. Class imbalance is addressed through moderated sampling that increases the probability of underrepresented relations without excessive rebalancing.

YafaRE (Melhem et al., 2026) YafaRE fine-tunes ArBERTv2 for relation classification and applies transductive self-training by pseudo-labeling the unlabeled test inputs to adapt the model to the test distribution, improving Micro-F1 from 70.20% to 91.92%. The team also explores two- and three-model ensembles with soft weighted voting, but neither outperforms the self-training approach.

5.4 Evaluation

For RE, participants submit a single `.txt` file through Codabench[4], containing one tab-separated `triple_id predicted_relation` entry per line. The number and order of entries must exactly match those of the released test file.

Each system predicts one of 41 labels for every entity pair: the 40 predefined relation types and *no_relation*. The *no_relation* label is derived from entity pairs for which no relation annotation is present, rather than being independently annotated as a relation type. Therefore, its frequency and prediction performance are not directly comparable to those of the 40 annotated relation types. Following the evaluation protocol adopted in relation classification benchmarks (Fan, 2023; Hendrickx et al., 2010; Zhou et al., 2020), instances with *no_relation* as the gold label are excluded from evaluation. Accordingly, F1 is computed only

[4] https://www.codabench.org/competitions/17338/

over instances whose gold label corresponds to one of the 40 predefined relation types.

5.5 Results and Discussion

Table 6 reports the official Relation Extraction results, ranked by micro-F1. Seven teams submitted valid runs. The table also includes the official baseline, based on NLI AraBERTv02 fine-tuned on the released training split.

Rank	Team	Precision	Recall	Micro-F1
1	U4RASD	95.55	**92.61**	**94.06**
2	NAMAA	96.80	90.38	93.48
3	ArmyForRE	94.78	91.91	93.32
4	Wasl	**97.16**	89.69	93.28
—	*Baseline*	—	—	93.09
5	McMasterNLP	96.56	89.60	92.95
6	ISNAD	92.74	92.36	92.55
7	YafaRE	96.19	88.01	91.92

Table 6: Official Relation Extraction results (%).

Among the submitted systems, the two LLM-based approaches, U4RASD and NAMAA, rank first and second with Micro-F1 scores of 94.06 and 93.48, respectively, followed by the two encoder-based ensemble systems, ArmyForRE (93.32) and Wasl (93.28); all four outperform the organizer baseline (93.09), while McMasterNLP (92.95), ISNAD (92.55), and YafaRE (91.92) fall below it.

Ontology-based constraints. Among the *LLM-based systems*, U4RASD (94.06) outperforms NAMAA (93.48). A notable methodological difference is that U4RASD incorporates hard ontology-based constraints to restrict candidate relations, whereas NAMAA does not use ontology constraints. This difference suggests that ontology-based constraints may contribute to U4RASD's stronger performance, although other differences between the two systems prevent their effect from being isolated. A similar pattern is observed among the *encoder-based* systems: ArmyForRE (93.32), Wasl (93.28), and McMasterNLP (92.95) all outperform ISNAD (92.55) and YafaRE (91.92). Among the top three, ArmyForRE applies hard ontology-based constraints, while Wasl and McMasterNLP incorporate a soft ontology-based type-constraint penalty during decoding; ISNAD and YafaRE use neither. The consistent association between ontology-based constraints — whether hard or soft — and higher performance within both model families suggests that such constraints may provide useful structural information for RE. However, this remains an observational finding, as the systems also differ in their architectures, training procedures, and other inference strategies.

Encoder- vs. LLM-based approaches: performance and computational considerations. Two of the four top-performing systems, ArmyForRE and Wasl, are encoder-based, while U4RASD's best submission (Yehia-RE) and NAMAA are LLM-based. NAMAA fine-tunes and ensembles three LoRA-adapted causal LLMs, whereas ArmyForRE and Wasl rely on much smaller BERT-scale encoders. Despite this difference in scale, the encoder-based systems remain competitive: ArmyForRE's seven-model soft ensemble (four XLM-RoBERTa-large seeds and three Arabic BERT-based models) trails NAMAA's three-model LLM ensemble by only 0.16 points, and Wasl's five-model encoder ensemble is 0.78 points behind U4RASD's best system. These results indicate that encoder-based ensembling, combined with ontology-based constraints, remains a competitive and computationally lighter alternative to LLM-based approaches.

Long-tail and semantically confusable relations remain challenging. Several systems employ imbalance-aware strategies, including focal loss (ArmyForRE) and frequency-based sampling (U4RASD, ISNAD), yet rare relations remain difficult to classify. Beyond class imbalance, Wasl's error analysis identifies persistent confusions between semantically close relations sharing the same argument-type signature, such as *birth_place* vs. *lives_in* and *capital_of* vs. *geopolitical_division*, which cannot be distinguished through ontology constraints alone. U4RASD similarly identifies difficulty with rare and semantically adjacent relations as a limitation.

6 Conclusion

KnowledgeGraphEval introduced a shared benchmark for Arabic Knowledge Graph construction through two subtasks: cross-domain NER (Adapt-NER) and Relation Extraction (RE). The results highlight persistent challenges in domain adaptation for Arabic NER, while RE, despite competitive results, still faces open challenges in disambiguating long-tail and semantically confusable relations. Future editions should address these limitations in both evaluation design and modeling.

Limitations

The study has two main limitations. First, the datasets used in the shared task are primarily representative of Modern Standard Arabic (MSA) and provide limited coverage of dialectal variation. Consequently, the findings may not fully reflect RE and NER performance across the broader spectrum of Arabic varieties.

Second, Subtask 2 (RE) formulates each instance around a single subject–object entity pair within a sentence. When a sentence contains multiple entity pairs or relations, each pair is treated as a separate classification instance. This formulation isolates individual relations and does not evaluate the joint identification and classification of multiple relations within a sentence. Future shared-task editions could consider a formulation that evaluates all candidate entity pairs within each sentence jointly.

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